Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? R1's Serial0/0/1 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129 2

Does R2 show up as an OSPF neighbor on R1? Yes 2

Does R2 show up as an OSPF neighbor on R3? No 2

What does this information tell you?

R3 is routing to 192.168.2.0/24 via the path through R1 with an accumulated OSPF cost of 129. R2 appears as an OSPF neighbor on R1 but not on R3, which indicates that R3 and R2 do not have a direct OSPF adjacency. 2

Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1 /

no passive-interface s0/0/1 /

h. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? R2's Serial0/0/1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

1 + 64 = 65 2

Is R2 listed as an OSPF neighbor to R3? Yes 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth get the same OSPF cost of 1. Changing it lets OSPF assign different costs proportionally for higher-speed links so the metric accurately reflects link speeds and path references.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For 192.168.3.0/24, the cost are calculated by the combination of R1's s0/0/1(or s0/0/1) (781) and R3's g0/0 (1), so the cost is 781 + 1 = 782. For 192.168.23.0/24, the cost are calculated by the combination of R1's s0/0/0(781) and R2's s0/0/0(64), or R1's s0/0/1(781) and R3's s0/0/0(64) so the cost is 781 + 64 = 845.

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 1562. As all the bandwidth are set to 128, every route has the same distance so the new cost is 2 hop: 781 + 781 = 1562.

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

As the cost of R1' s0/0/1 is increased to 1565, the path to the 192.168.3.0/24 network through R3 now has a higher cost than that of through R2. OSPF always selects the lowest-cost path, so the path (through R2) is selected.

4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   When the OSPF protocol is used, the OSPF router ID uniquely identifies each router within an OSPF domain and is used in LSAs, SPF calculation, and election processes. If the router ID is not explicitly set, it may be chosen unpredictably which can change after reboots or interface changes, causing instability.

   4

2. Why is the DR/BDR election process not a concern in this lab?

   This lab uses point-to-point serial links. On P2P links there is no DR/BDR, so DR/BDR election process is not a concern.

   4

3. Why would you want to set an OSPF interface to passive?

   This prevents OSPF from sending Hellos and forming neighbor adjacencies on that interface while still advertising the connected network in OSPF. This improve security, reduce CPU and adjacency overhead, and prevent unnecessary OSPF traffic

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF's area-based hierarchy, which can reduces routing table and CPU load and improves scalability.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   router ospf 1

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   As RIP AD = 120, OSPF AD = 110, EIGRP AD = 90, EIGRP has the lowest administrative distance so the route from EIGRP will be used for the routing decision.

   4